COMPUTER ENGINEERING

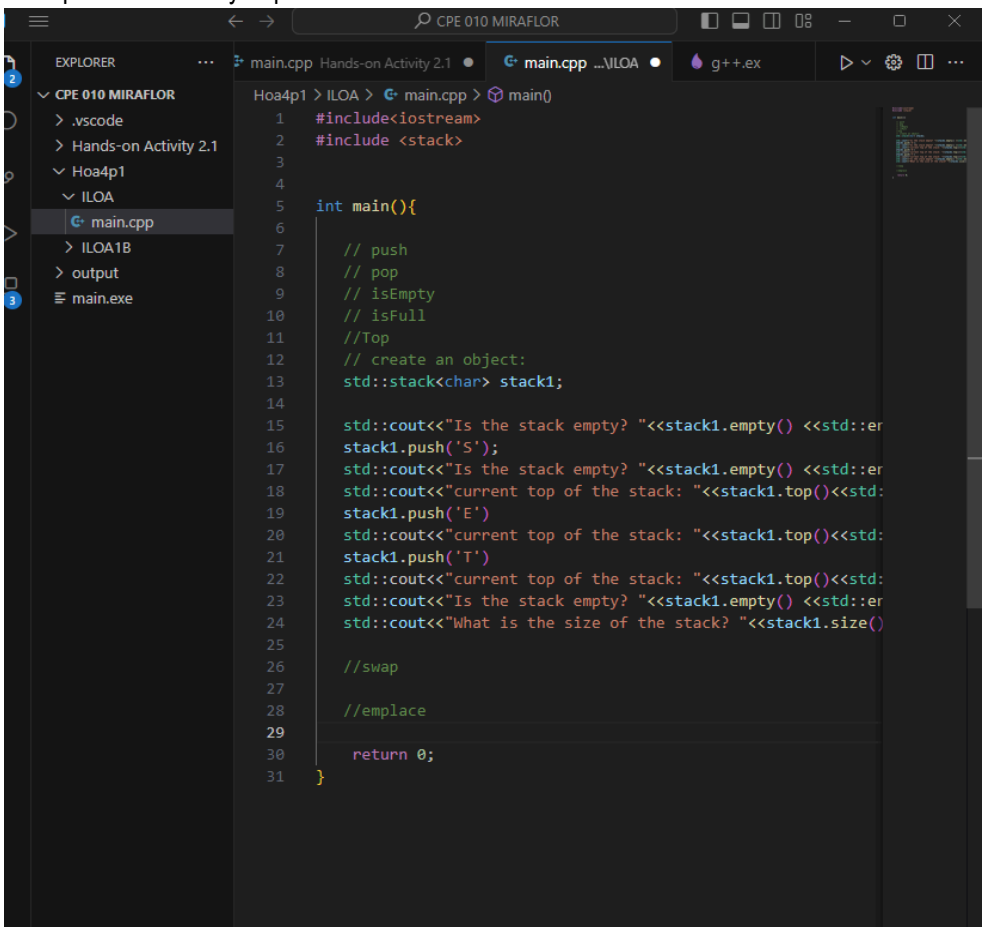
QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S4
DATE PERFORMED:	29/07/2026	**DATE SUBMITTED:**	29/07/2026
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ACTIVITY NO. ___ [ACTIVITY NAME]

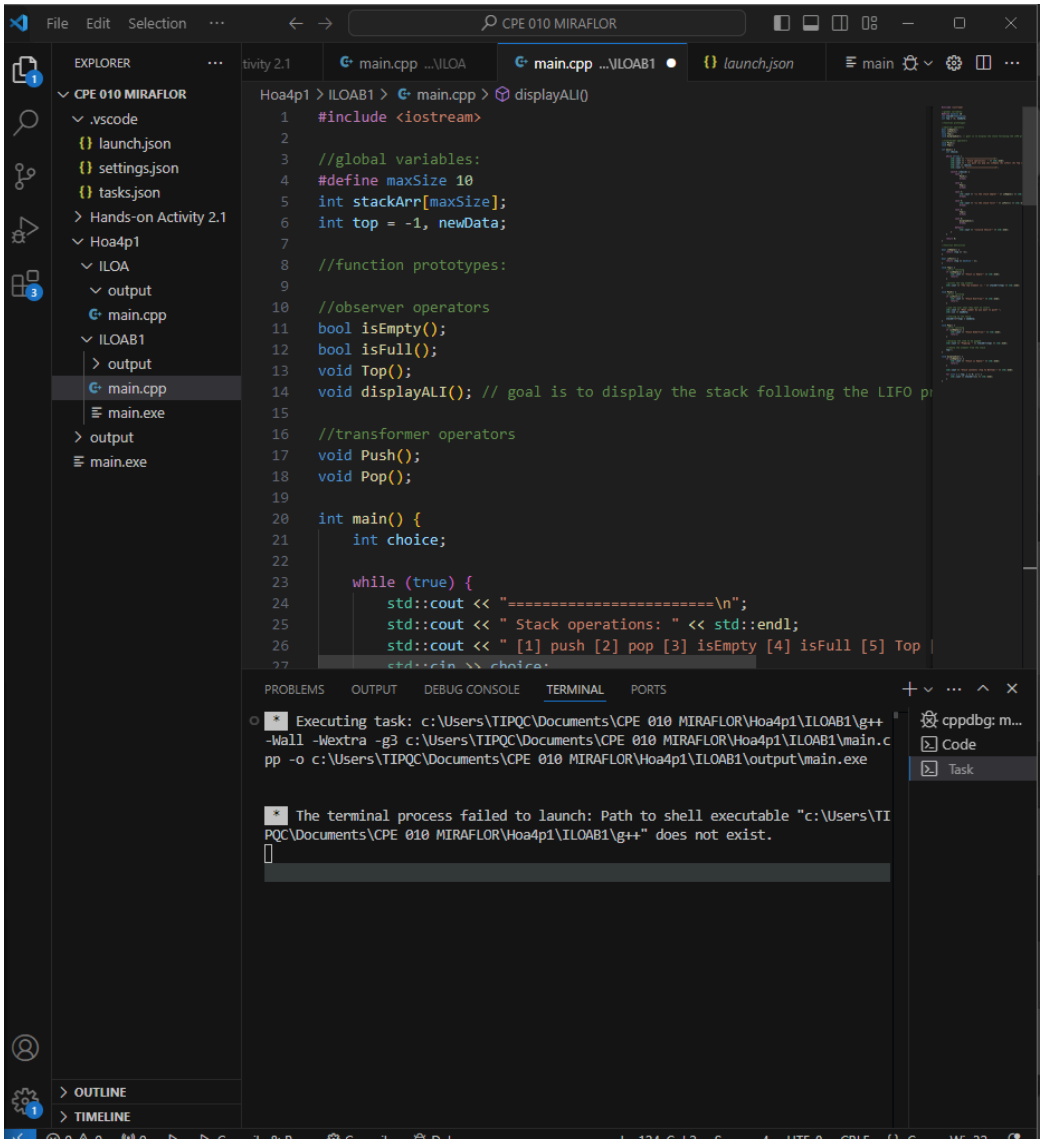
1. PROCEDURE OUTPUT

What procedures did you perform?



```
38      // emplace
39      stack1.emplace('H');
40      std::cout << "\nAfter emplace('H')" << std::endl;
41      std::cout << "Current top of the stack: " << stack1.top() << std::en
42      std::cout << "Size: " << stack1.size() << std::endl;
43
44      // swap
45      std::stack<char> stack2;
46      stack2.push('A');
47      stack2.push('B');
48
49      std::cout << "\nBefore swap()" << std::endl;
50      std::cout << "Top of stack1: " << stack1.top() << std::endl;
51      std::cout << "Top of stack2: " << stack2.top() << std::endl;
52
53      stack1.swap(stack2);
54
55      std::cout << "\nAfter
56      std::cout << "Top of
57      std::cout << "Top of stack2: " << stack2.top() << std::endl;
58
59      return 0;
60  }
```

(const char [16])"Top of stack2: "


File Edit Selection
CPE 010 MIRAFLOR
EXPLORER
CPE 010 MIRAFLOR
.vscode
launch.json
settings.json
tasks.json
Hands-on Activity 2.1
Hoa4p1
ILOA
output
main.cpp
ILOAB1
output
main.cpp
main.exe
output
main.exe
OUTLINE
TIMELINE
main.cpp ...\ILOA
main.cpp ...\ILOAB1
launch.json
Hoa4p1 > ILOAB1 > main.cpp > displayALI()
#include <iostream>
//global variables:
#define maxSize 10
int stackArr[maxSize];
int top = -1, newData;
//function prototypes:
//observer operators
bool isEmpty();
bool isFull();
void Top();
void displayALI(); // goal is to display the stack following the LIFO p
//transformer operators
void Push();
void Pop();
int main() {
int choice;
while (true) {
std::cout << "=======================\n";
std::cout << " Stack operations: " << std::endl;
std::cout << " [1] push [2] pop [3] isEmpty [4] isFull [5] Top
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Executing task: c:\Users\TIPQC\Documents\CPE 010 MIRAFLOR\Hoa4p1\ILOAB1\g++ -Wall -Wextra -g3 c:\Users\TIPQC\Documents\CPE 010 MIRAFLOR\Hoa4p1\ILOAB1\main.cpp -o c:\Users\TIPQC\Documents\CPE 010 MIRAFLOR\Hoa4p1\ILOAB1\output\main.exe
The terminal process failed to launch: Path to shell executable "c:\Users\TIPQC\Documents\CPE 010 MIRAFLOR\Hoa4p1\ILOAB1\g++" does not exist.
cppdbg: m...
Code
Task

```
113 void displayALI() {
114     if (isEmpty()) {
115         std::cout << "Stack is Empty!" << std::endl;
116         return;
117     }
118
119     std::cout << "Stack contents (Top to Bottom):" << std::endl;
120
121     for (int i = top; i >= 0; i--) {
122         std::cout << stackArr[i] << std::endl;
123     }
124 }
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

========================
 Stack operations:
 [1] push [2] pop [3] isEmpty [4] isFull [5] Top [6] Display
5
========================
Stack is Empty!
========================
 Stack operations:
 [1] push [2] pop [3] isEmpty [4] isFull [5] Top [6] Display
7
========================
invalid Choice!
========================
 Stack operations:
 [1] push [2] pop [3] isEmpty [4] isFull [5] Top [6] Display
6
========================
Stack is Empty!
========================
 Stack operations:
 [1] push [2] pop [3] isEmpty [4] isFull [5] Top [6] Display
```

ANALYSIS

- The program implements a stack data structure using an array and follows the LIFO (Last In, First Out) principle. It uses the Top variable to track the current position and provides functions for pushing, popping, checking if the stack is empty or full, viewing the top element, and displaying the stack contents. The program correctly handles stack overflow and underflow conditions, and its operations are efficient with most functions having O(1) time complexity. However, it can be improved by reducing the use of global variables and adding more flexible features such as dynamic sizing and an exit option.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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